

Xi HE

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Employment

The University of Texas at Dallas (UTD)

May. 2025 to Present

Research Associate, Department of Physics, Supervisor: Prof. Bei Zeng

The Chinese University of Hong Kong (CUHK)

Mar. 2022 to May. 2025

Postdoctoral Fellow, Department of Physics, Supervisor: Prof. Renbao Liu

Tencent Holdings Limited

Jun. 2020 to Aug. 2020

Intern, Tencent Quantum Laboratory, Supervisor: Jonathan Allcock

Education

Ph.D. University of Electronic Science and Technology of China (UESTC)

Sept. 2017 to Jun. 2021

Institute of Fundamental and Frontier Sciences (IFFS)

- **Supervisor:** Prof. Xiaoting Wang
- **Major:** Physics
- **GPA:** 3.93/4

M.Sc. University of Electronic Science and Technology of China

Sept. 2015 to Jun. 2017

School of Physics and Electronics

- **Supervisor:** Prof. Qingyuan Wang
- **Major:** Radio Physics
- **GPA:** 3.83/4

B.Eng. University of Electronic Science and Technology of China

Sept. 2010 to Jun. 2014

School of Physics and Electronics

- **Major:** Electronic Engineering

Research Interests

- **Quantum machine learning:** Design machine learning algorithms on quantum devices to achieve quantum advantages with theoretical or numerical guarantees.
- **Deep learning enhanced quantum information extraction:** Using deep neural networks to learn the evolution or data of quantum systems.
- **Quantum cross-domain learning; Quantum reinforcement learning.**

Honors & Awards

[1] **Academic Scholarship for Doctoral Student**

Oct. 2020

[2] **Academic Scholarship for Doctoral Student**

Oct. 2019

[4] **Academic Scholarship for Doctoral Student**

Oct. 2018

[5] **Freshman Scholarship for Doctoral Student**

Sept. 2017

[6] **Excellent Graduate Student**

Dec. 2016

[7] **Academic Scholarship for Master Student**

Oct. 2016

[8] **Freshman Scholarship for Master Student**

Sept. 2015

[9] **People's Scholarship**

Dec. 2011

Academic Service

Journal review: Quantum Science and Technology, Machine Learning: Science and Technology, Quantum Information Processing, Journal of Neural Engineering, Physica Scripta, New Journal of Physics.

Research Grants Awarded

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| [1] National Natural Science Foundation of China (General Program)
Research on segmentation methods of abdominal CT images with limited resources
Project No: 62271296 | Jan. 2023 to Dec. 2026
Participant |
| [2] Natural Science Basic Research Program of Shaanxi
Quantization, application and interpretability of deep transfer learning algorithms
Project No: 2022JQ-018 | Jan. 2022 to Dec. 2024
Principal Investigator |
| [3] National Key Research and Development Program of China
Quantum machine learning and quantum simulation
Project No: 2018YFA0306703 | May. 2018 to Apr. 2023
Participant |

Publications

Journal papers

- [1] Hengyuan Hu, Minglei Yan, Jietao Jiang, Ankui Huang, Sicheng Cai, Linxuan Lan, Kewei Ye, Danlei Chen, Kewen Tang, Qin Zuo, Yun Zeng, Wei Tang, Junheng Fu, Chenglu Jiang, Yong Wang, Zhenhua Yan, **Xi He**, Liang Qiao, Yang Zhao. A state-of-the-art review on biomass-derived carbon materials for supercapacitor applications: From precursor selection to design optimization. *Science of The Total Environment*, 2024, 912, 169141.
- [2] **Xi He**, Feiyu Du, Mingyuan Xue, Xiaogang Du, Tao Lei, AK Nandi. Quantum classifiers for domain adaptation. *Quantum Information Processing*, 2023, 22(2): 105.
- [3] **Xi He**. Quantum subspace alignment for domain adaptation. *Physical Review A*, 2020, 102(6): 062403.
- [4] **Xi He**. Quantum correlation alignment for unsupervised domain adaptation. *Physical Review A*, 2020, 102(3): 032410.
- [5] **Xi He**, Li Sun, Chufan Lyu, Xiaoting Wang. Quantum locally linear embedding for nonlinear dimensionality reduction. *Quantum Information Processing*, 2020, 19: 1-21.

Conference papers

- [1] Feiyu Du, **Xi He**. Quantum joint distribution adaptation on the universal quantum computer. *IEEE 6th International Conference on Pattern Recognition and Artificial Intelligence (PRAI)*, 2023.
- [2] Tao Lei, Rui Sun, Xuan Wang, Yingbo Wang, **Xi He**, Asoke Nandi. CiT-Net: Convolutional neural networks hand in hand with vision transformers for medical image segmentation. *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence (IJCAI-23)*, 2023.
- [3] **Xi He**, Feiyu Du. Quantum kernel subspace alignment for unsupervised domain adaptation. *Proceedings of the 2nd Asia Conference on Algorithms, Computing and Machine Learning*, 2023.
- [4] Xinzhe Geng, Tao Lei, Qi Chen, Jian Su, **Xi He**, Qi Wang, Asoke K Nandi. Global evolution neural network for segmentation of remote sensing images. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022.
- [5] Li Sun, **Xi He**, Chenglong You, Chufan Lyu, Bo Li, Seth Lloyd, Xiaoting Wang. Exponentially enhanced quantum metrology using resources of linear complexity. *The 20th Asian Quantum Information Science Conference (AQISC2020)*, 2020.
- [6] Xiaokai Hou, **Xi He**, Chufan Lv, Dingding Wen, Xiaoting Wang. Quantum linear regression with regularization. *APS March Meeting*, 2019.

- [7] **Xi He**, Li Sun, Xiaokai Hou, Xiaoting Wang. Quantum manifold learning algorithms for dimensionality reduction. APS March Meeting, 2019.
- [8] Peng Zhao, Qingyuan Wang, Fan Zhang, **Xi He**, A novel T-junction waveguide power divider with anti-phases and broad bandwidth. 2017 IEEE AP-S Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting (IEEE AP-S/URSI 2017), 2017

Patents

- [1] Qingyuan Wang, **Xi He**, Peng Zhao. Wideband ridge transmission line coupling structure. Patent No.2017204807470; Grant No.CN206947488U.
- [2] Qingyuan Wang, Tianyu Zhang, **Xi He**. Back hole coupled ridge transmission line filter structure. Patent No.2017204812816; Grant No.CN206742465U.
- [3] Qingyuan Wang, **Xi He**, Peng Zhao. Orthogonal mode adapter with mirror symmetrical structure. Patent No.2017203415797; Grant No.CN206657853U.

Preprints

- [1] **Xi He**, Sirui Lu, Bei Zeng. Co-designing quantum codes with transversal diagonal gates via multi-agent systems. arXiv: 2510.20728, (2025).
- [2] Arunaday Gupta, Baisong Sun, **Xi He**, Bei Zeng. Variance geometry of exact Pauli-detecting codes: continuous landscapes beyond stabilizers. arXiv: 2604.21800, (2026).
- [3] **Xi He**, Feiyu Du, Xiaohan Yu, Yang Zhao, Tao Lei. Distribution alignment based transfer fusion frameworks on quantum devices for seeking quantum advantages. arXiv: 2411.01822, (2024).
- [4] **Xi He**, Chufan Lyu, Min-Hsiu Hsieh, Xiaoting Wang. Quantum transfer component analysis for domain adaptation. arXiv: 1912.09113, (2019).

Invited/Plenary/Keynote Presentations in Conferences

- [1] Xi He, IEEE 6th International Conference on Pattern Recognition and Artificial Intelligence (PRAI2023) (18-20 Aug, 2023 online). Quantum Joint Distribution Adaptation on the Universal Quantum Computer.
- [2] Xi He, Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence (IJCAI-23) (19-25, Aug, 2023, Macao SAR, China). CiT-Net: Convolutional Neural Networks Hand in Hand with Vision Transformers for Medical Image Segmentation.
- [3] Xi He, Proceedings of the 2nd Asia Conference on Algorithms, Computing and Machine Learning (CACML2023) (Mar. 17-19, 2023, online). Quantum kernel subspace alignment for unsupervised domain adaptation.
- [4] Xi He, APS March Meeting 2019 (APS2019) (Mar. 4–8, 2019, Boston, Massachusetts, USA), Quantum Manifold Learning Algorithms for Dimensionality Reduction.

Referee List

Prof. Bei Zeng	SNSM, UTD
Mail: Bei.Zeng@utdallas.edu;	
Prof. Xiaoting Wang	IFFS, UESTC
Mail: xiaoting@uestc.edu.cn;	
Prof. Weichuan Zhang	EIAI, SUST
Mail: zwc2003@163.com;	
Prof. Wen-Long Ma	IS, CAS
Mail: wenlongma@semi.ac.cn;	

Prof. Xiaohan Yu

Mail: xiaohan.yu@mq.edu.au;

SC, MQU